

SESSION 17

Payment Gateway Integration

Topics Covered: Paytm/Stripe/PayPal integration, Payment initiation, Transaction callback

Activities: Create 'Book Appointment' -> 'Pay Now' flow

Task 1: Set up basic Flask app and create /pay route displaying payment form for IPL match tickets (name, email, amount).

Explanation & Approach: *Minimal Flask application rendering ticket purchase payment initiation page.*

```
from flask import Flask, render_template_string, request

app = Flask(__name__)

HTML_FORM = '''
<h2>IPL Match Ticket Booking</h2>
<form action="/paytm-initiate" method="POST">
    <input type="text" name="name" placeholder="Full Name" required><br><br>
    <input type="email" name="email" placeholder="Email" required><br><br>
    <input type="number" name="amount" value="500" readonly><br><br>
    <button type="submit">Pay Now with Paytm</button>
</form>
'''

@app.route('/pay')
def pay_route():
    return render_template_string(HTML_FORM)

if __name__ == '__main__':
    app.run(debug=True)
```

Task 2: Integrate Paytm payment gateway sandbox in Flask app so form submission redirects to Paytm test payment page.

Explanation & Approach: *Generates Paytm checksum and posts HTML form redirect payload.*

```
# Paytm Initiate Route
@app.route('/paytm-initiate', methods=['POST'])
def paytm_initiate():
    paytm_params = {
        "MID": "YOUR_MID_HERE",
        "ORDER_ID": "ORDER_1001",
        "CUST_ID": request.form['email'],
        "TXN_AMOUNT": request.form['amount'],
        "CHANNEL_ID": "WEB",
        "INDUSTRY_TYPE_ID": "Retail",
        "WEBSITE": "WEBSTAGING",
        "CALLBACK_URL": "http://127.0.0.1:5000/payment-callback",
    }
```

```
# Checksum generation using PaytmChecksum library
# checksum = PaytmChecksum.generateSignature(paytm_params, "YOUR_MERCHANT_KEY")
return f"Redirecting to Paytm Sandbox with Order ID: {paytm_params['ORDER_ID']}..."
```

Task 3: Handle transaction callback from Paytm by creating /payment-callback route that verifies transaction status.

Explanation & Approach: *Verifies checksum signature and handles RESPCODE status.*

```
@app.route('/payment-callback', methods=['POST'])
def payment_callback():
    callback_data = request.form.to_dict()
    # verify signature logic
    if callback_data.get('RESPCODE') == '01':
        return "<h1>Payment Successful! Ticket Confirmed.</h1>"
    else:
        return "<h1>Payment Failed or Cancelled.</h1>", 400
```

Task 4: Simulate 'Pay Now' experience for Zomato-style food order: modify Flask app for Stripe test payment.

Explanation & Approach: *Creates Stripe PaymentIntent using official Python SDK.*

```
import stripe
from flask import Flask, jsonify, request

stripe.api_key = "sk_test_YOUR_STRIPE_TEST_KEY"

@app.route('/create-stripe-payment-intent', methods=['POST'])
def create_payment():
    try:
        data = request.json
        intent = stripe.PaymentIntent.create(
            amount=int(data['amount']) * 100, # Amount in paise/cents
            currency='inr',
            payment_method_types=['card'],
        )
        return jsonify({'clientSecret': intent['client_secret']})
    except Exception as e:
        return jsonify({'error': str(e)}), 400
```

Task 5: Use ChatGPT to generate Python code for integrating PayPal payment processing in Flask route, adapt and comment.

Explanation & Approach: *Integrates PayPal REST SDK v1/v2 checkout order creation.*

```
# PayPal Flask Integration (Prompt & Code)
# Prompt: "Generate Flask code for PayPal REST SDK checkout flow"
import paypalrestsdk
```

```
paypalrestsdk.configure({
    "mode": "sandbox", # sandbox or live
    "client_id": "YOUR_PAYPAL_CLIENT_ID",
    "client_secret": "YOUR_PAYPAL_CLIENT_SECRET"
})

@app.route('/paypal-pay', methods=['POST'])
def paypal_pay():
    payment = paypalrestsdk.Payment({
        "intent": "sale",
        "payer": {"payment_method": "paypal"},
        "redirect_urls": {
            "return_url": "http://localhost:5000/paypal-success",
            "cancel_url": "http://localhost:5000/paypal-cancel"
        },
        "transactions": [{
            "amount": {"total": "10.00", "currency": "USD"},
            "description": "IPL Ticket Booking Payment"
        }]
    })

    if payment.create():
        for link in payment.links:
            if link.rel == "approval_url":
                return redirect(link.href)
    return "Error creating PayPal payment", 500
```